

EE 101: Basic Electronics

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Magnetic circuits

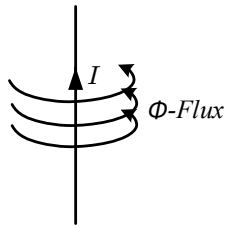
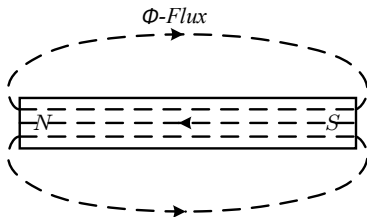
Introduction

- Electromagnetic (EM) is the branch of Electrical Engineering (or Physics) that deals with the analysis and application of electric and magnetic fields.
- The principles of EM are applied in various disciplines, such as electrical machines, electromechanical energy conversion, radar technology, remote sensing, satellite communications, wireless power transfer (WPT), bioelectromagnetic, electromagnetic interference and compatibility, etc.
- EM devices include electric motors and generators, antennas, radars, microwave ovens, WPT systems, superconductors, MRI, and electrocardiograms. Design of these devices requires a thorough knowledge of the laws and principles of EM.
- Electrical Engineers who specialize in EM are needed in microwave industries, communication industries and electromagnetic research laboratories.

Basic Parameters

- μ - Permeability (H/m , henry/meter)
- μ_0 - Permeability of the free space ($4\pi \times 10^{-7} H/m$)
- ϕ - Flux (Wb , weber)
- B - Flux density (Wb/m^2 , weber/square meter)
- H - Magnetic Field intensity (A/m , ampere/meter)
- F - MMF, Magneto motive force (AT , ampere-turns)
- R - Reluctance
- ρ - Permeance

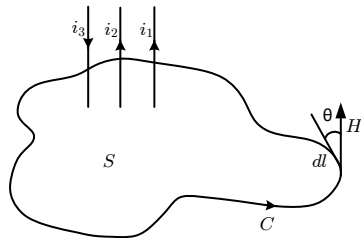
Magnetic Field



Ampere's Law

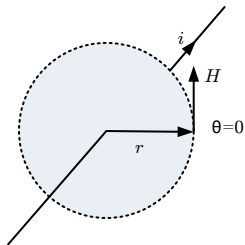
$$\oint H \cdot dl = \Sigma i = i_1 + i_2 - i_3$$

$$\oint H dl \cos \theta = \Sigma i$$



$$\oint H \cdot dl = i \Rightarrow H 2\pi r = i$$

$$\Rightarrow H = \frac{i}{2\pi r}$$



Permeability

$$\frac{\text{Flux density at C}}{\text{Magnetic field strength at C}} = \frac{B}{H} = \mu$$

$\mu (= \mu_0 \mu_r) \rightarrow$ Characteristics of the medium called Permeability

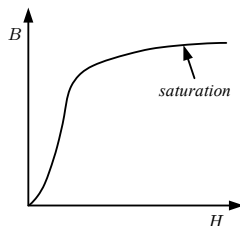
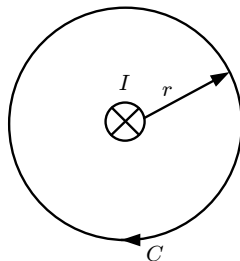
$\mu_0 \rightarrow$ Permeability of free space, $4\pi \times 10^{-7}$ H/m

$\mu_r \rightarrow$ Relative permeability of the medium

Air, Cu, Al, insulators $\rightarrow \mu_r = 1$

Ferromagnetic materials: $\mu_r \rightarrow 2000 - 6000$

High μ_r value indicates that a small current can produce large flux density in the machines

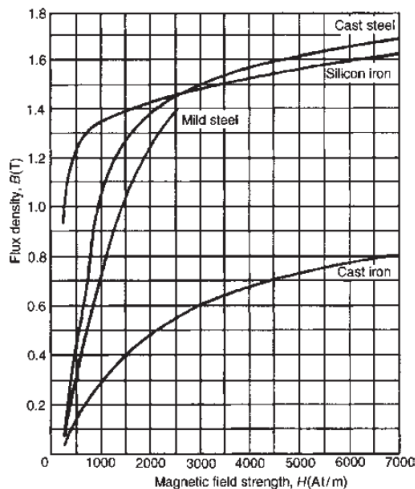


Permeability (cont..)

Lower $H \rightarrow$ linear relation of B and H .

Higher $H \rightarrow$ non-linear relation of B and H

$H \rightarrow$ Magnetic material is operating in saturation region



MMF

$$\oint H \cdot dl = Ni \Rightarrow H 2\pi r = Ni$$

$$\Rightarrow H = \frac{Ni}{2\pi r} = \frac{Ni}{l_c}$$

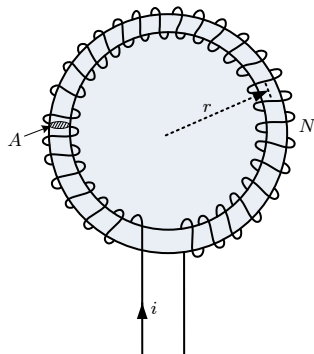
$$B = \frac{\mu Ni}{l_c}$$

$l_c \rightarrow$ length of magnetic path/ core

$A \rightarrow$ cross-sectional area of core

The quantity Ni is called magneto motive force (mmf) which is denoted as $\mathbf{F}$. Its unit is ampere turns (AT).

Magnetic field intensity is related to magnetic flux density.



Reluctance

$$B = \mu H = \frac{Ni}{l_c/\mu}$$
$$\phi = BA = \frac{Ni}{l_c/\mu A}$$

If $R = l_c/\mu A$, where R is the reluctance of a magnetic circuit.

The flux in terms of mmf and the reluctance can be written as

$$\phi = \frac{Ni}{R} = \frac{mmf}{Reluctance}$$

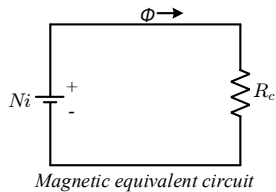
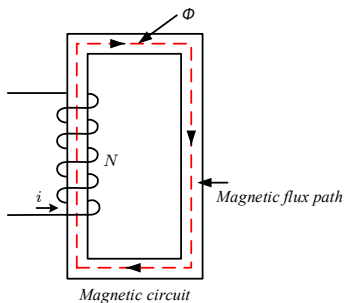
Permeance, $P = 1/R$

Magnetic Equivalent Circuit

$$mmf \leftrightarrow emf$$

$$\phi \leftrightarrow I$$

$$R \leftrightarrow \mathbb{R}$$



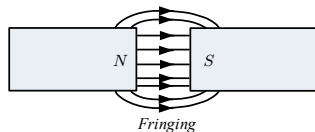
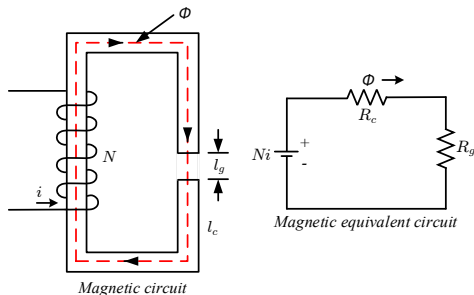
Magnetic Circuit with Air Gap

The reluctance of the core: $R_c = \frac{l_c}{\mu A_c}$

The reluctance of air-gap: $R_g = \frac{l_g}{\mu_o A_g}$

The common flux: $\phi = \frac{Ni}{R_c + R_g}$

Fringing: Magnetic flux lines bulge outward called fringing of flux. Small air-gap $\rightarrow$ fringing is neglected



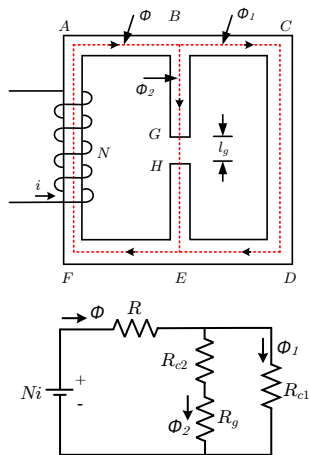
Parallel Magnetic Circuits

The reluctance of the core:

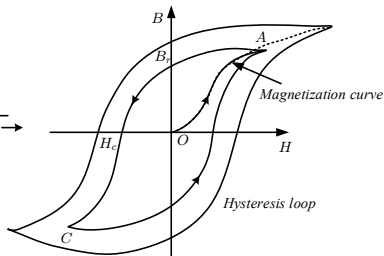
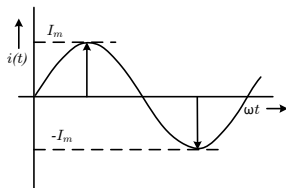
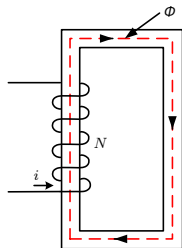
$$R_c = \frac{l_{BAFE}}{\mu A_c} \quad R_{c1} = \frac{l_{BCDE}}{\mu A_c} \quad R_{c2} = \frac{l_{BGHE} - l_g}{\mu A_c}$$

The reluctance of air-gap: $R_g = \frac{l_g}{\mu_o A_g}$

The flux: $\Phi = \frac{Ni}{R_c + R_{c1} \parallel (R_{c2} + R_g)}$



Hysteresis Loss



B-H loop is nonlinear and multi valued

$$P_H = k_h B_{max}^m f$$

Here, P_H is hysteresis loss density (W/m^3)

Eddy Current Loss

Empirical formula: $P_e = K_e f^2 B_{max}^2$

where K_e is constant

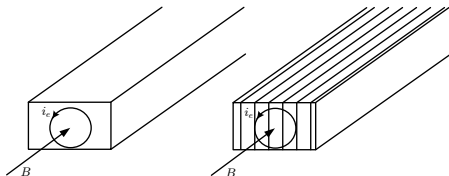
Here, P_e is eddy current loss density (W/m^3)

Reduction:

(a) High resistivity core material (addition of silicon (4%) to iron).

(b) Laminated core $\rightarrow$ 0.5 to 5 mm in machine

Core loss: $P_c = P_h + P_e$



Magnetically Coupled Circuits



Wireless charging of phone

Inductance

A coil wound on a magnetic core can be represented by an ideal circuit element, called inductance. Inductance is defined as the flux linkage of the coil per unit current. Flux linkage is given as

$$\lambda = N\phi$$

$$L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{NBA}{i}$$

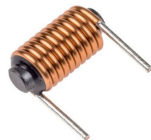
where λ is the flux linkage, B is the flux density and A is the cross-sectional area. If H is the magnetic field intensity and l_c is the length of the core, then

$$Hl_c = Ni$$

$$\Rightarrow i = \frac{Hl_c}{N}$$

$$L = \frac{NBA}{i} = \frac{N\mu HA}{i} = \frac{N\mu HA}{Hl_c/N}$$

$$L = \frac{N^2}{l_c/\mu A} = \frac{N^2}{R}$$



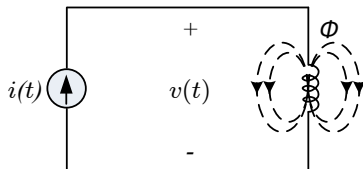
Self Inductance

$$v = N \frac{d\phi}{dt}$$

$$v = N \frac{d\phi}{di} \frac{di}{dt}$$

$$v = L \frac{di}{dt}$$

$$L = N \frac{d\phi}{di}$$



Magnetic flux produced by a single coil with N turns

Mutual Inductance

When two inductors (or coils) are in a close proximity to each other, the magnetic flux caused by current in one coil links with the other coil, thereby inducing voltage in the latter. This phenomenon is known as *mutual inductance*.

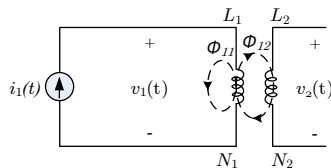
$$\phi_1 = \phi_{11} + \phi_{12}$$

$$v_1 = N_1 \frac{d\phi_1}{dt} \quad v_2 = N_2 \frac{d\phi_{12}}{dt}$$

$$v_1 = N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} = L_1 \frac{di_1}{dt} \quad v_2 = N_2 \frac{d\phi_{12}}{di_1} \frac{di_1}{dt} = M_{21} \frac{di_1}{dt}$$

$$M_{21} = N_2 \frac{d\phi_{12}}{di_1}$$

$$v_2 = M_{21} \frac{di_1}{dt}$$



Mutual inductance of coil 2 with respect to coil 1

M_{21} is known as the *mutual inductance* of coil 2 with respect to coil 1.

Mutual Inductance (cont..)

$$\phi_2 = \phi_{21} + \phi_{22}$$

$$v_2 = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d\phi_2}{di_2} \frac{di_2}{dt} = L_2 \frac{di_2}{dt}$$

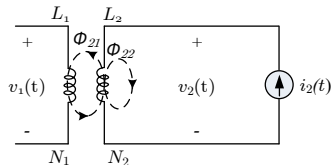
$$v_1 = N_1 \frac{d\phi_{21}}{dt} = N_1 \frac{d\phi_{21}}{di_2} \frac{di_2}{dt} = M_{12} \frac{di_2}{dt}$$

$$M_{12} = N_1 \frac{d\phi_{21}}{di_2}$$

$$v_1 = M_{12} \frac{di_2}{dt}$$

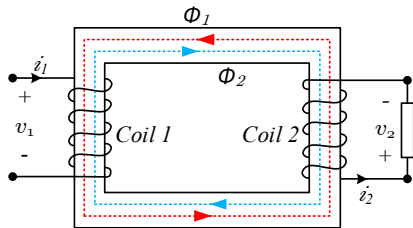
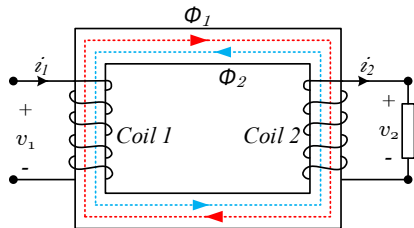
To be proved later:

$$M_{12} = M_{21} = M$$



Mutual inductance of coil 1 with respect to coil 2

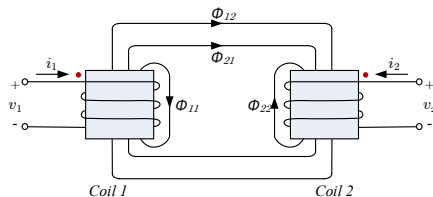
Polarity of Induced Voltage



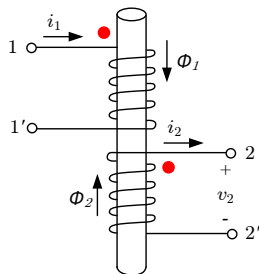
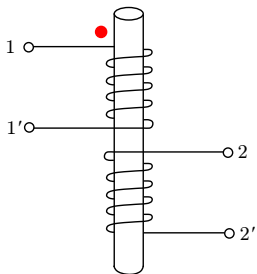
Dot Convention

If a current **enters** the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **positive** at the dotted terminal of the second coil.

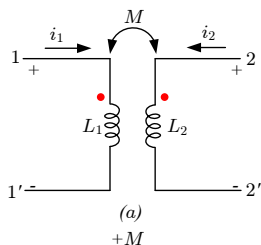
If a current **leaves** the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **negative** at the dotted terminal of the second coil.



Dot Convention (cont..)



Dot Convention (Cont..)

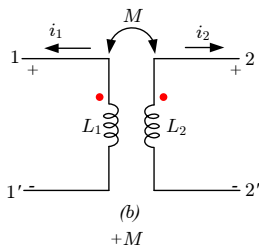


$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

$$\bar{V}_1 = j\omega L_1 \bar{I}_1 + j\omega M \bar{I}_2$$

$$\bar{V}_2 = j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1$$

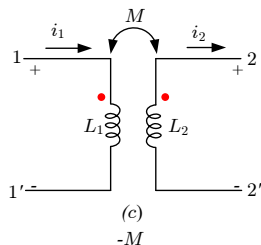


$$v_1 = -L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = -L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

$$\bar{V}_1 = -j\omega L_1 \bar{I}_1 - j\omega M \bar{I}_2$$

$$\bar{V}_2 = -j\omega L_2 \bar{I}_2 - j\omega M \bar{I}_1$$



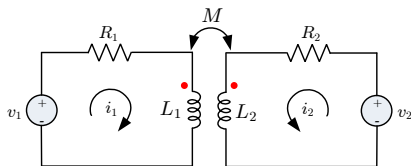
$$v_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = -L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

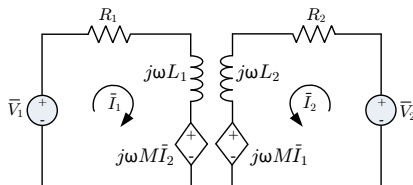
$$\bar{V}_1 = j\omega L_1 \bar{I}_1 - j\omega M \bar{I}_2$$

$$\bar{V}_2 = -j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1$$

Dot Convention (Cont..)

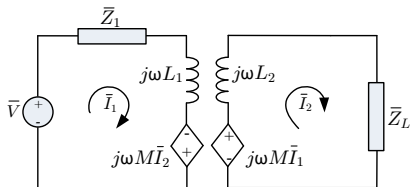
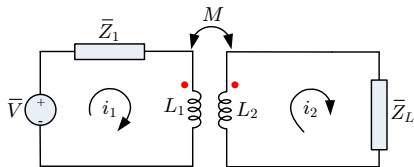


$$v_1 = i_1 R_1 + L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$
$$v_2 = i_2 R_2 + L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$



$$\bar{V}_1 = (R_1 + j\omega L_1) \bar{I}_1 + j\omega M \bar{I}_2$$
$$\bar{V}_2 = j\omega M \bar{I}_1 + (R_2 + j\omega L_2) \bar{I}_2$$

Dot Convention (Cont..)



$$\bar{V} = (\bar{Z}_1 + j\omega L_1)\bar{I}_1 - j\omega M \bar{I}_2$$

$$0 = -j\omega M \bar{I}_1 + (\bar{Z}_L + j\omega L_2)\bar{I}_2$$

Equivalent Inductance

Fig.(a): $v(t) - L_1 \frac{di}{dt} - L_2 \frac{di}{dt} = 0$

$$\rightarrow v(t) = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

$$\therefore L_{eq} = L_1 + L_2$$

Fig.(b): Two inductors are magnetically coupled:

$$v(t) - L_1 \frac{di}{dt} + M \frac{di}{dt} - L_2 \frac{di}{dt} + M \frac{di}{dt} = 0$$

$$\rightarrow v(t) = L_1 \frac{di}{dt} - M \frac{di}{dt} + L_2 \frac{di}{dt} - M \frac{di}{dt}$$

$$= (L_1 + L_2 - 2M) \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

$$\therefore L_{eq} = L_1 + L_2 - 2M$$

Fig.(c): $v(t) - L_1 \frac{di}{dt} - M \frac{di}{dt} - L_2 \frac{di}{dt} - M \frac{di}{dt} = 0$

$$\therefore L_{eq} = L_1 + L_2 + 2M$$

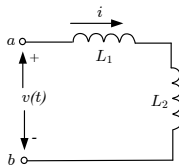


Fig. (a)

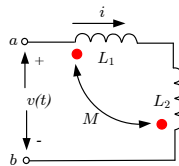


Fig. (b)

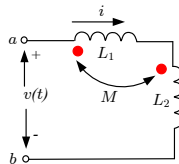
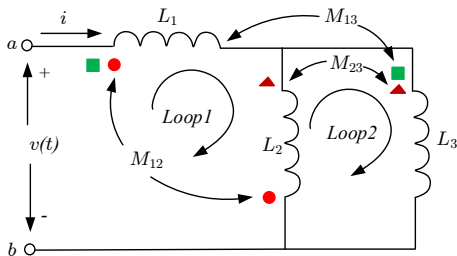


Fig. (c)

Equivalent Inductance (Cont..)

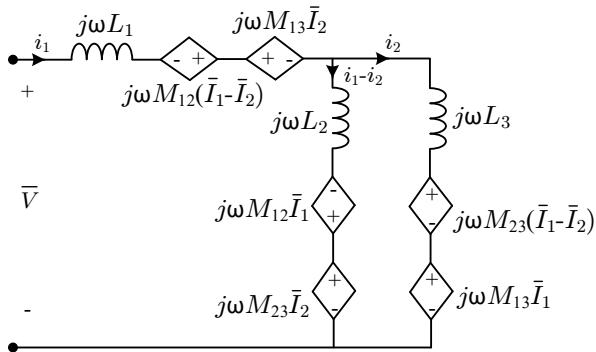


Loop1:

$$v(t) - L_1 \frac{di_1}{dt} + M_{12} \frac{d(i_1 - i_2)}{dt} - M_{13} \frac{di_2}{dt} - L_2 \frac{d(i_1 - i_2)}{dt} + M_{12} \frac{di_1}{dt} - M_{23} \frac{di_2}{dt} = 0$$

Loop2:

$$-L_3 \frac{di_2}{dt} - M_{23} \frac{d(i_1 - i_2)}{dt} - M_{13} \frac{di_1}{dt} + M_{23} \frac{di_2}{dt} - M_{12} \frac{di_1}{dt} + L_2 \frac{d(i_1 - i_2)}{dt} = 0$$



Loop1:

$$\bar{V} - j\omega L_1 \bar{I}_1 + j\omega M_{12}(\bar{I}_1 - \bar{I}_2) - j\omega M_{13} \bar{I}_2 - j\omega L_2(\bar{I}_1 - \bar{I}_2) + j\omega M_{12} \bar{I}_1 - j\omega M_{23} \bar{I}_2 = 0$$

Loop2:

$$-j\omega L_3 \bar{I}_2 - j\omega M_{23}(\bar{I}_1 - \bar{I}_2) - j\omega M_{13} \bar{I}_1 + j\omega M_{23} \bar{I}_2 - j\omega M_{12} \bar{I}_1 + j\omega L_2(\bar{I}_1 - \bar{I}_2) = 0$$

Energy Stored in Inductor

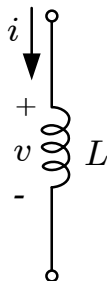
$$p = vi = \left(L \frac{di}{dt} \right) i$$

The energy stored is

$$\begin{aligned} w &= \int_{-\infty}^t p \, dt = \int_{-\infty}^t \left(L \frac{di}{dt} \right) i \, dt \\ &= L \int_{-\infty}^t i \, di = \frac{1}{2} L i^2(t) - \frac{1}{2} L i^2(-\infty) \end{aligned}$$

Since $i(-\infty) = 0$,

$$w = \frac{1}{2} L i^2$$



Energy in a Coupled Circuit

Step 1: $i_2 = 0$, i_1 increased from 0 to I_1

$$p_1(t) = v_1 i_1 = i_1 L_1 \frac{di_1}{dt}, \quad v_2 i_2 = 0$$

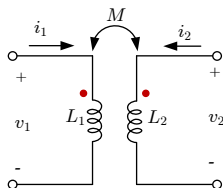
$$w_1 = \int p_1 dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2} L_1 I_1^2$$

Step 2: $i_1 = I_1$, i_2 increased from 0 to I_2

$$p_2(t) = i_1 M_{12} \frac{di_2}{dt} + i_2 v_2 = I_1 M_{12} \frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt}$$

$$\begin{aligned} w_2 &= \int p_2 dt = M_{12} I_1 \int_0^{I_2} di_2 + L_2 \int_0^{I_2} i_2 di_2 \\ &= M_{12} I_1 I_2 + \frac{1}{2} L_2 I_2^2 \end{aligned}$$

$$w = w_1 + w_2 = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{12} I_1 I_2$$



Energy in a Coupled Circuit (cont..)

If sequence of steps changed i.e.

Step 1: $i_1 = 0$, i_2 increased from 0 to I_2

Step 2: $i_2 = 0$, i_1 increased from 0 to I_1

$$w = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M_{21} I_1 I_2$$

$$M_{12} = M_{21} = M$$

$$w = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M I_1 I_2$$

If current direction is reversed in one of the inductors,

$$w = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 - M I_1 I_2$$

Generalizing the above, magnetic energy stored in coupled inductors is given by

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$

Coefficient of coupling

$$\frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 - Mi_1i_2 \geq 0$$

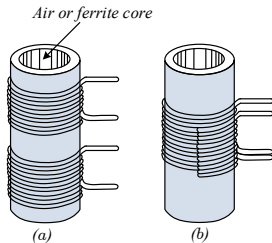
$$\frac{1}{2}(i_1\sqrt{L_1} - i_2\sqrt{L_2})^2 + i_1i_2(\sqrt{L_1L_2} - M) \geq 0$$

This is true for any values of i_1, i_2, L_1, L_2 and M . This gives,

$$\begin{aligned}\sqrt{L_1L_2} - M &\geq 0 \\ \Rightarrow M &\leq \sqrt{L_1L_2}\end{aligned}$$

$$k = \frac{M}{\sqrt{L_1L_2}} \quad \boxed{M = k\sqrt{L_1L_2}}$$

The **coupling coefficient** k is a measure of the magnetic coupling between two coils; $0 \leq k \leq 1$.



Windings: (a) loosely occupied, (b) tightly coupled; cutaway view demonstrates both windings.

Thank You!